



Lecture 7

Robustness to Effect Heterogeneity in GLMs

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Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

Why might this assumption fail?

- The nature of the conditional relationship is not appropriately captured by the model's form.
- The model imposes conditionality that is not really there.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What kinds of inferential instability can result?

- Misleading conclusions about the average effect.
- Improper estimation of effect sizes/directions (e.g., Simpson's paradox).
- Poor predictions.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What tools exist to evaluate robustness?

- Interaction models
- Smooth interactions with GAMs

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What are the limits of these tools?

- In essence, all variables are moderators - which to investigate?
- Evidence for moderation is often low power.
 - Interaction effects are typically estimated much less precisely than average effects.
 - Reliable estimation requires sufficient support across combinations of x and the moderator.

Transition from Linear to GLM

Consider a linear model: $y = b_0 + b_1x + b_2z + e$

- The effect of x is $\frac{\partial y}{\partial x} = b_1$

In the non-linear model: $E(y|x, z) = g^{-1}(b_0 + b_1x + b_2z) = g^{-1}(\eta)$

- The effect of x is

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x}$$

Common Links and First Derivatives

Model	Response Function	Effect of x
Linear	$E(Y) = \eta$	β_x
Binary	$p = F(\eta)$	$\beta_x f(\eta)$
Log-Link	$\mu = e^\eta$	$\beta_x \mu$
Ordered	$P(Y = k) = F(\tau_k - \eta) - F(\tau_{k-1} - \eta)$	$\beta_x [f(\tau_{k-1} - \eta) - f(\tau_k - \eta)]$
Multinomial Logit	$p_k = \frac{e^{\eta_k}}{\sum_j e^{\eta_j}}$	$p_k (\beta_{xk} - \sum_j p_j \beta_{xj})$

where

- $F(\cdot)$ is a cumulative distribution function (CDF)
- $f(\cdot) = F'(\cdot)$ is the corresponding probability density function (PDF)

Binary Model Example

Consider the binary regression model:

$$\log \left(\frac{p}{1-p} \right) = b_0 + b_1x + b_2z$$

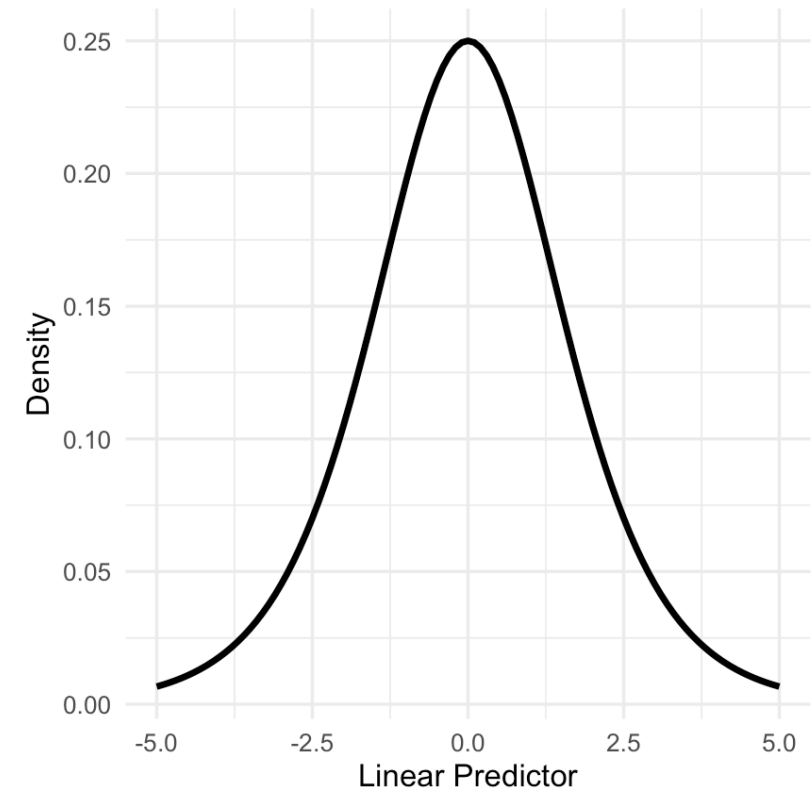
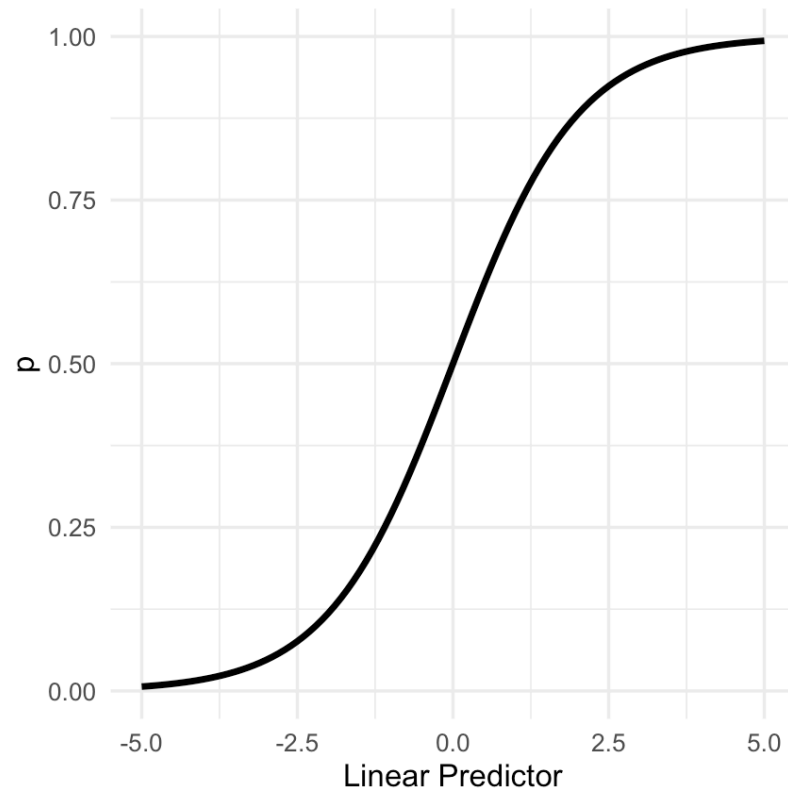
The effect of x is:

$$\frac{\partial E(y|x, z)}{\partial x} = f(\eta)b_1 = p(1-p)b_1$$

Because p depends on all predictors, the effect of x changes across values of x and z .

Effect Visualization

The effect is biggest when $p = 0.5$ and gets smaller in either direction.



Link vs Product Term Conditionality

Mechanical conditionality that comes from the non-linear link:

- *can* dampen or amplify the coefficient of x as a function of z .
- *cannot* allow a coefficient to change signs.

Including a product term:

- *can* permit sign flips based on different values of the conditioning variable.
- *can* reduce the level of conditionality between x and z in some circumstances.

Link vs Product Term Conditionality

If $E(y|x, z) = g^{-1}(b_0 + b_1x + b_2z)$

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} b_1$$

If $E(y|x, z) = g^{-1}(b_0 + b_1x + b_2z + b_3xz)$

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} (b_1 + b_3z)$$

Conditionality in GLMs

If the model induces conditionality by definition, how do we think about moderation?

1. Just as before - generally, an investigation about conditionality should be theoretically interesting.
 - At this stage, you should not go "hunting for" interactions.
2. Unlike before, the mechanical model building exercise does not lead to clear, immediate answers.
 - Whether the product term is significant and the extent of the moderation are less related here.
3. Whether a product term belongs in the model depends on whether it generates substantively important conditionality beyond the mechanical conditionality induced by the link function.

Evaluating Conditionality

Variable Types	Marginal Effect of x	Interaction Quantity
x, z both continuous	$\frac{\partial E(Y)}{\partial x}$	$\frac{\partial}{\partial z} \frac{\partial E(Y)}{\partial x}$
x continuous, z binary	$\frac{\partial E(Y)}{\partial x}$	$\Delta_z \left(\frac{\partial E(Y)}{\partial x} \right)$
x, z both binary	$\Delta_x E(Y)$	$\Delta_z (\Delta_x E(Y))$

Evaluating Conditionality

Variable Types	Marginal Effect of x	Interaction Quantity
x, z both continuous	$\frac{\partial E(Y)}{\partial x}$	$\frac{\partial}{\partial z} \frac{\partial E(Y)}{\partial x}$
x continuous, z binary	$\frac{\partial E(Y)}{\partial x}$	$\Delta_z \left(\frac{\partial E(Y)}{\partial x} \right)$
x, z both binary	$\Delta_x E(Y)$	$\Delta_z (\Delta_x E(Y))$

Finite Difference Approximation for 1 - interquartile change in marginal effects:

$$\frac{\partial}{\partial z} \frac{\partial E(Y)}{\partial x} \approx \frac{\frac{\partial E(Y)}{\partial x} \Big|_{z=P_{75}} - \frac{\partial E(Y)}{\partial x} \Big|_{z=P_{25}}}{P_{75} - P_{25}}$$

Example: ANES Voting Democrat

```
anes <- rio::import("../data/anes1992.dta")
anesu <- anes |>
  mutate(retnat = rio::factorize(retnat),
         educ = rio::factorize(educ)) |>
  drop_na(votedem, retnat, pid, age, male,
         educ, black, south)
add_mod <- glm(votedem ~ retnat + pid + age + male + educ +
              black + south, data=anesu, family=binomial)
tidy(add_mod)
```

```
## # A tibble: 9 × 5
##   term          estimate std.error statistic  p.value
##   <chr>         <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.512     0.673      0.760  4.47e- 1
## 2 retnatsame     1.33      0.594      2.24  2.50e- 2
## 3 retnatworse    1.65      0.570      2.89  3.89e- 3
## 4 pid           -0.916    0.0557    -16.4  1.06e-60
## 5 age            0.0131   0.00566     2.32  2.05e- 2
## 6 male           -0.292    0.180     -1.62  1.04e- 1
## 7 educ           0.251    0.104      2.42  1.57e- 2
## 8 black          2.15     0.477      4.50  6.64e- 6
## 9 south          0.0129   0.221      0.0584 9.53e- 1
```

Two Categorical Variables: *retnat* and *black*

```
econ_comps <- avg_comparisons(add_mod,
  datagrid(male = c(0,1), grid_type="counterfactual"),
  variables = list(retnat = c("better", "worse")),
  by = "male")
econ_comps
```

```
##
## male Estimate Std. Error      z Pr(>|z|)    S  2.5 % 97.5 %
##    0      0.203      0.0666 3.04  0.00235 8.7 0.0720  0.333
##    1      0.199      0.0638 3.12  0.00180 9.1 0.0741  0.324
##
## Term: retnat
## Type: response
## Comparison: worse - better
```

```
econ_comps |> hypotheses(~pairwise)
```

```
##
## Hypothesis Estimate Std. Error      z Pr(>|z|)    S  2.5 % 97.5 %
## (b2) - (b1) -0.00331      0.00364 -0.909  0.363 1.5 -0.010
```



Example: ANES Voting Democrat

Two Categorical Variables: *retnat* and *male*

```
prod_mod <- glm(votedem ~ male*retnat + pid + age + black + e
               south, data=anesu, family=binomial)
tidy(prod_mod)
```

```
## # A tibble: 11 × 5
##   term                estimate std.error statistic  p.value
##   <chr>              <dbl>      <dbl>      <dbl>    <dbl>
## 1 (Intercept)        -1.30        1.44      -0.899  3.69e- 1
## 2 male                2.04        1.54       1.33   1.84e- 1
## 3 retnatsame          3.02        1.44       2.10   3.56e- 2
## 4 retnatworse         3.49        1.41       2.47   1.34e- 2
## 5 pid               -0.916       0.0558    -16.4   1.21e-60
## 6 age                 0.0130       0.00568    2.30   2.17e- 2
## 7 black               2.22        0.484      4.59   4.41e- 6
## 8 educ                0.259       0.104      2.48   1.31e- 2
## 9 south               0.0178       0.223     0.0801  9.36e- 1
## 10 male:retnatsame    -2.15        1.59      -1.36   1.75e- 1
## 11 male:retnatworse   -2.44        1.55      -1.57   1.16e- 1
```

```
econ_comps2 <- avg_comparisons(prod_mod,
                                datagrid(male = c(0,1), grid_type="counterfactual"),
                                variables = list(retnat = c("better", "worse")),
                                by = "male")
```

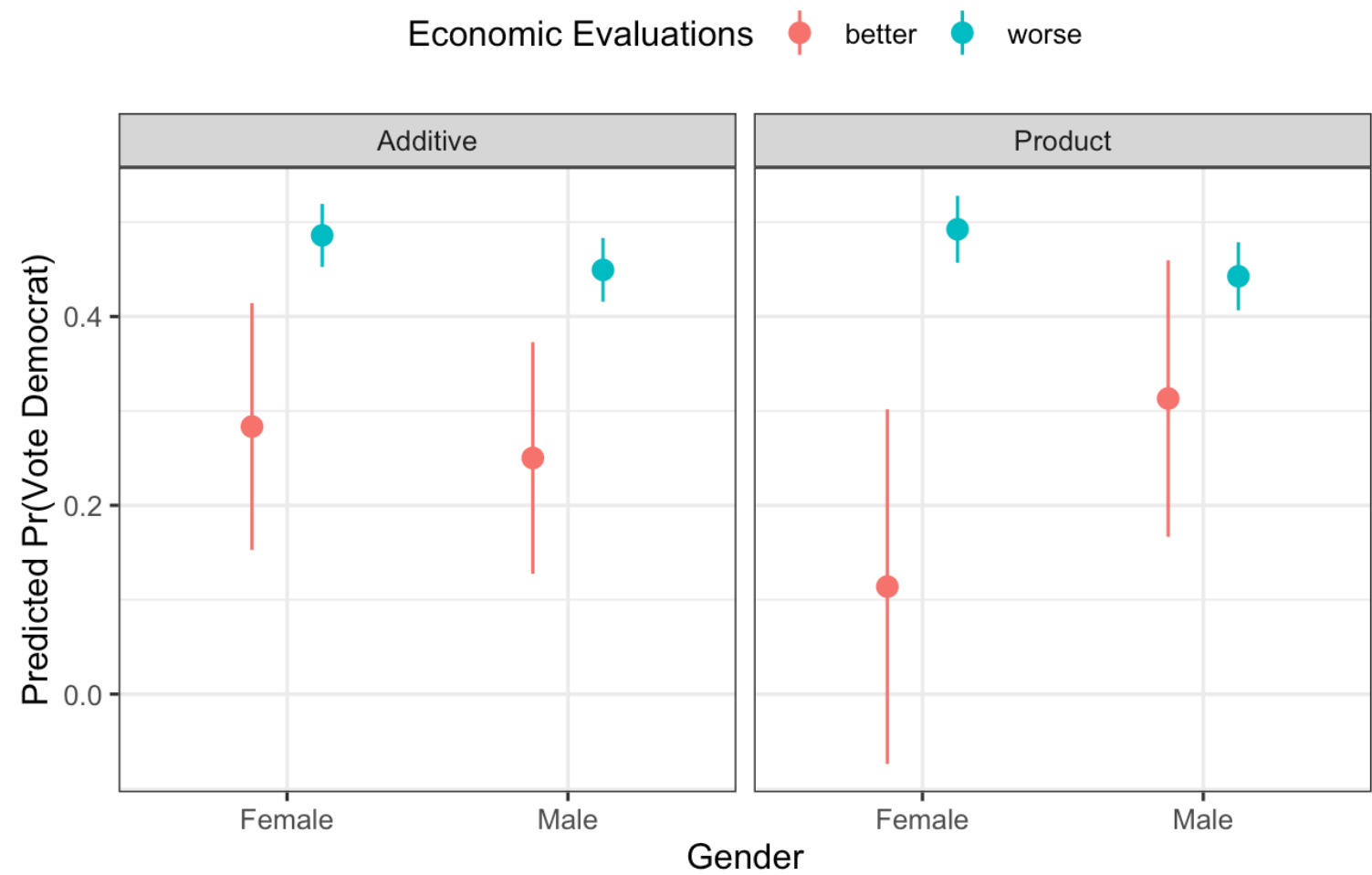
```
econ_comps2
```

```
##
##   male Estimate Std. Error    z Pr(>|z|)    S  2.5 % 97.5 %
##   0      0.379    0.0976 3.88  <0.001 13.2  0.1873  0.57
##   1      0.129    0.0770 1.68  0.0925  3.4 -0.0214  0.28
##
## Term: retnat
## Type: response
## Comparison: worse - better
```

```
econ_comps2 |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error    z Pr(>|z|)    S  2.5 % 9
##   (b2) - (b1)   -0.249    0.124 -2    0.0452  4.5 -0.493 -0.
```


Effects:





Example: ANES Voting Democrat

```
add_mod <- glm(votedem ~ retnat + pid + age + male + educ +
              black + south, data=anesu, family=binomial)
tidy(add_mod)
```

```
## # A tibble: 9 × 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.512     0.673      0.760  4.47e- 1
## 2 retnatsame     1.33      0.594      2.24  2.50e- 2
## 3 retnatworse    1.65      0.570      2.89  3.89e- 3
## 4 pid           -0.916    0.0557    -16.4  1.06e-60
## 5 age            0.0131   0.00566     2.32  2.05e- 2
## 6 male          -0.292    0.180     -1.62  1.04e- 1
## 7 educ           0.251    0.104     2.42  1.57e- 2
## 8 black          2.15     0.477     4.50  6.64e- 6
## 9 south          0.0129   0.221     0.0584 9.53e- 1
```

One Categorical, One Numeric: *pid* and *black*

```
pid_slopes <- avg_slopes(add_mod,
  datagrid(black = c(0,1), grid_type="counterfactual"),
  variables = "pid",
  by = "black")

pid_slopes
```

```
##
##   black Estimate Std. Error      z Pr(>|z|)      S 2.5 % 97.5 %
##   0    -0.119    0.00213  -55.8  <0.001    Inf -0.123 -0.117
##   1    -0.109    0.00595  -18.3  <0.001  246.3 -0.121 -0.097
##
## Term: pid
## Type: response
## Comparison: dY/dX
```

```
pid_slopes |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error      z Pr(>|z|)      S 2.5 % 97.5 %
##   (b2) - (b1)  0.00997    0.00516   1.93  0.0533  4.2 -0.00014 0.0201
```



Example: ANES Voting Democrat

```
prod_mod <- glm(votedem ~ male + retnat + pid*black + age + e
               south, data=anesu, family=binomial)
tidy(prod_mod)
```

```
## # A tibble: 10 × 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>    <dbl>    <dbl>
## 1 (Intercept)    0.571      0.673     0.848 3.96e- 1
## 2 male          -0.294      0.180    -1.63 1.02e- 1
## 3 retnatsame     1.31       0.591     2.21 2.69e- 2
## 4 retnatworse    1.64       0.567     2.90 3.78e- 3
## 5 pid           -0.930     0.0577   -16.1 2.21e-58
## 6 black          1.28       0.847     1.51 1.32e- 1
## 7 age            0.0129     0.00567   2.28 2.27e- 2
## 8 educ           0.255      0.104     2.45 1.42e- 2
## 9 south          0.00259     0.221     0.0117 9.91e- 1
## 10 pid:black     0.254      0.219     1.16 2.46e- 1
```

One Categorical, One Numeric: *pid* and *black*

```
pid_slopes2 <- avg_slopes(prod_mod,
  datagrid(black = c(0,1), grid_type="counterfactual"),
  variables = "pid",
  by = "black")

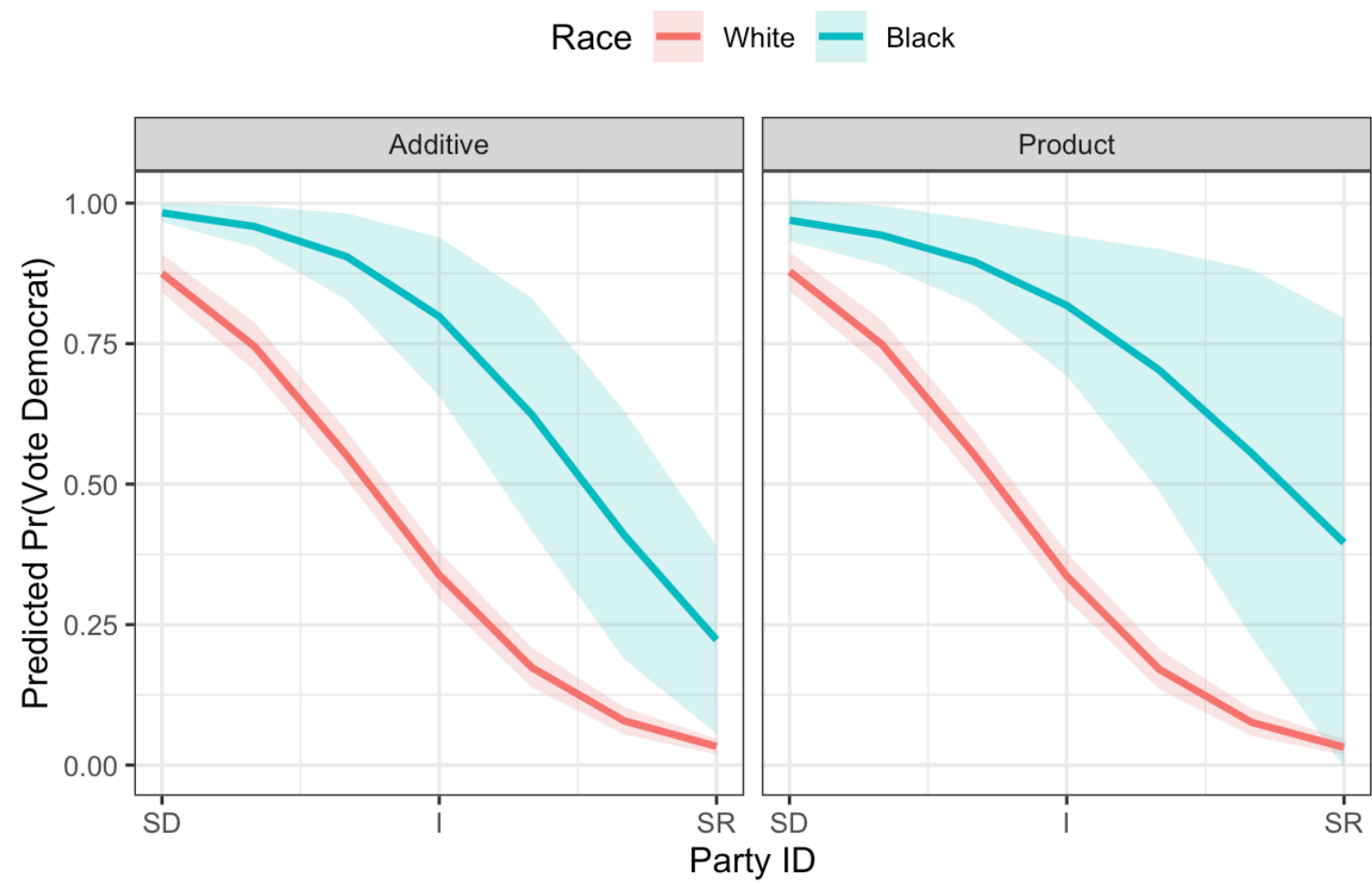
pid_slopes2
```

```
##
##   black Estimate Std. Error      z Pr(>|z|)    S 2.5 % 97.5
##    0   -0.1193    0.00209  -56.94 < 0.001 Inf -0.123 -0.11
##    1   -0.0879    0.02674   -3.29 0.00101 9.9 -0.140 -0.03
##
## Term: pid
## Type: response
## Comparison: dY/dX
```

```
pid_slopes2 |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error      z Pr(>|z|)    S 2.5 % 9
##   (b2) - (b1)   0.0314    0.0267  1.18    0.24 2.1 -0.021 0
```

Effects:



Example: ANES Voting Democrat

Two Numeric: *pid* and *age*

```
add_mod <- glm(votedem ~ retnat + pid + age + male + educ +
              black + south, data=anesu, family=binomial)
tidy(add_mod)
```

```
## # A tibble: 9 × 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)  0.512    0.673     0.760  4.47e- 1
## 2 retnatsame   1.33    0.594     2.24  2.50e- 2
## 3 retnatworse  1.65    0.570     2.89  3.89e- 3
## 4 pid        -0.916   0.0557    -16.4  1.06e-60
## 5 age         0.0131  0.00566     2.32  2.05e- 2
## 6 male       -0.292   0.180     -1.62  1.04e- 1
## 7 educ        0.251   0.104     2.42  1.57e- 2
## 8 black       2.15    0.477     4.50  6.64e- 6
## 9 south       0.0129  0.221     0.0584 9.53e- 1
```

```
pid_slopes <- avg_slopes(add_mod,
  datagrid(age = c(20, 60), grid_type="counterfactual"),
  variables = "pid",
  by = "age")
```

```
pid_slopes
```

```
##
##   age Estimate Std. Error      z Pr(>|z|)    S   2.5 % 97.5 %
##   20   -0.112    0.00263  -42.6  <0.001 Inf -0.117 -0.107
##   60   -0.112    0.00198  -56.3  <0.001 Inf -0.116 -0.108
##
## Term: pid
## Type: response
## Comparison: dY/dX
```

```
pid_slopes |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error      z Pr(>|z|)    S   2.5 % 97.5 %
##   (b2) - (b1) 0.000116    0.000764  0.152    0.879 0.2 -0.0013
```

Example: ANES Voting Democrat

Two Numeric: *pid* and *age*

```
prod_mod <- glm(votedem ~ male + retnat + pid*age + black + e
               south, data=anesu, family=binomial)
tidy(prod_mod)
```

```
## # A tibble: 10 × 5
##   term      estimate std.error statistic    p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept) -0.286    0.868    -0.329    0.742
## 2 male        -0.305    0.180    -1.69     0.0904
## 3 retnatsame   1.30     0.590     2.20     0.0275
## 4 retnatworse 1.61     0.567     2.83     0.00464
## 5 pid         -0.689    0.164    -4.20     0.0000268
## 6 age          0.0303   0.0133    2.28     0.0226
## 7 black        2.19     0.479     4.57     0.00000495
## 8 educ         0.274    0.105     2.60     0.00938
## 9 south        0.00976   0.221     0.0442    0.965
## 10 pid:age     -0.00487  0.00339   -1.43     0.152
```

```
pid_slopes2 <- avg_slopes(prod_mod,
  datagrid(age = c(20, 60), grid_type="counterfactual"),
  variables = "pid",
  by = "age")
```

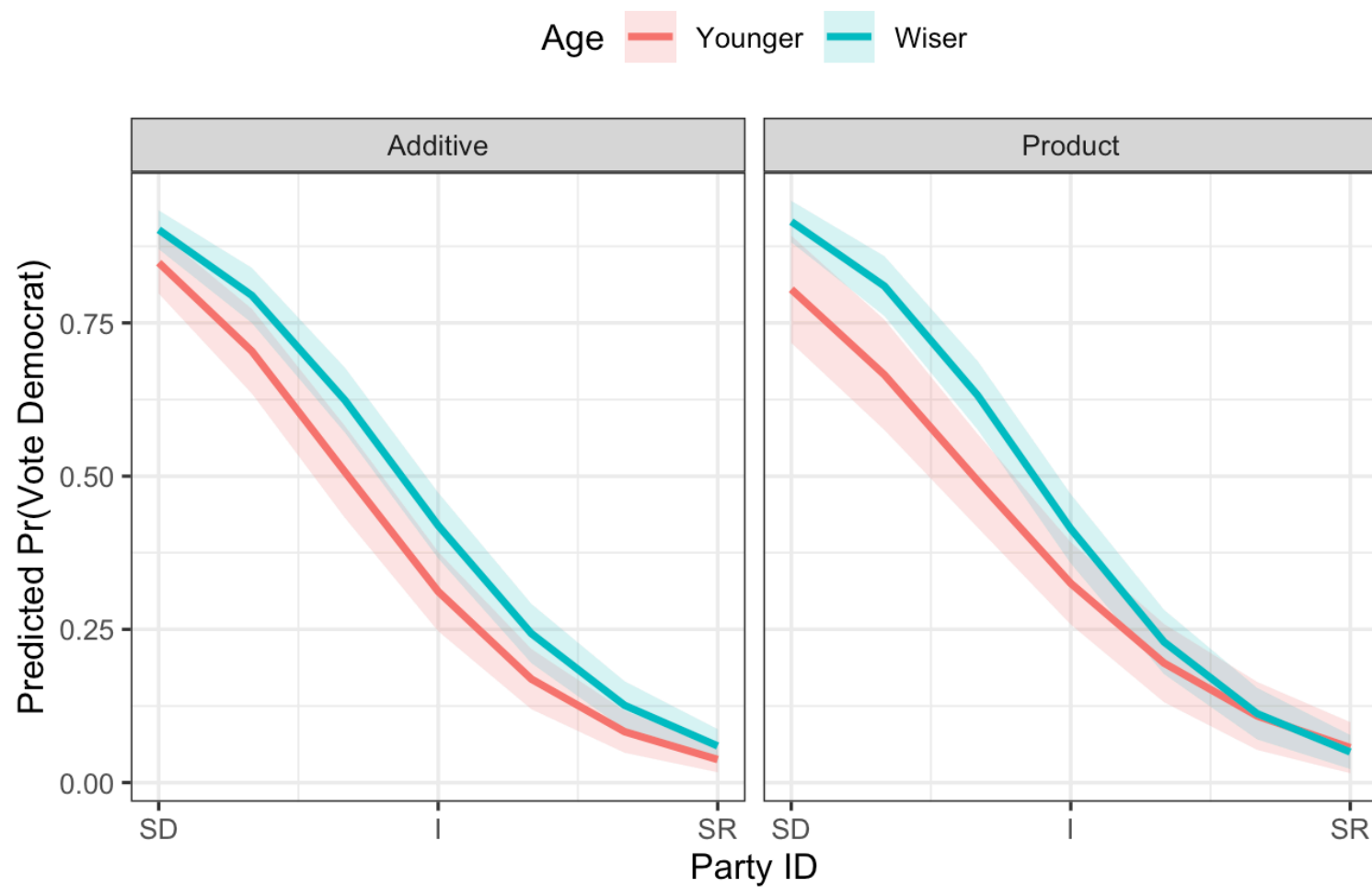
```
pid_slopes2
```

```
##
##   age Estimate Std. Error      z Pr(>|z|)      S 2.5 % 97.5 %
##   20   -0.106   0.00571  -18.6  <0.001 254.3 -0.117 -0.095
##   60   -0.113   0.00188  -60.0  <0.001   Inf -0.117 -0.109
##
## Term: pid
## Type: response
## Comparison: dY/dX
```

```
pid_slopes2 |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error      z Pr(>|z|)      S 2.5 % 97.5 %
##   (b2) - (b1) -0.00673   0.00575  -1.17   0.242 2.0 -0.018 0.004
```

Effects:





Product Term can Reduce Moderation

```
g_add <- glm(y ~ x + z, data=ex_dat, family=binomial)
comps_add <- avg_comparisons(g_add,
                             datagrid(z = c(0,1), grid_type = "counterfact",
                             variables = list(x = c(.25, .75)),
                             by = "z")
```

comps_add

```
##
##   z Estimate Std. Error    z Pr(>|z|)    S  2.5 % 97.5 %
##   0   0.1685     0.0326 5.17   <0.001 22.0 0.1046 0.2323
##   1   0.0597     0.0131 4.54   <0.001 17.4 0.0339 0.0855
##
## Term: x
## Type: response
## Comparison: 0.75 - 0.25
```

```
comps_add |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error    z Pr(>|z|)    S  2.5
##   (b2) - (b1)  -0.109     0.0229 -4.76   <0.001 19.0 -0.15
```

```
g_prod <- glm(y ~ x * z, data=ex_dat, family=binomial)
comps_prod <- avg_comparisons(g_prod,
                              datagrid(z = c(0,1), grid_type = "counterfact",
                              variables = list(x = c(.25, .75)),
                              by = "z")
```

comps_prod

```
##
##   z Estimate Std. Error    z Pr(>|z|)    S  2.5 % 97.5 %
##   0   0.119     0.0377 3.16   0.00159 9.3 0.0451 0.193
##   1   0.105     0.0216 4.88   < 0.001 19.8 0.0631 0.148
##
## Term: x
## Type: response
## Comparison: 0.75 - 0.25
```

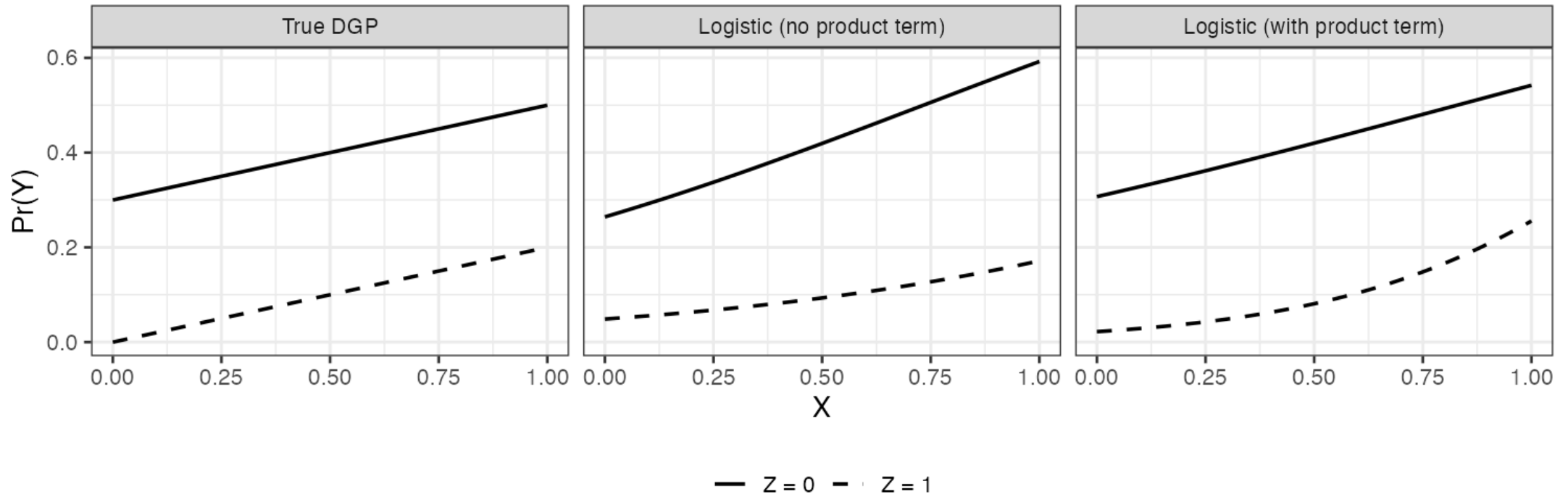
```
comps_prod |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error    z Pr(>|z|)    S  2.5
##   (b2) - (b1)  -0.0135     0.0435 -0.312   0.755 0.4 -0.09
```


Reducing Moderation - What's Happening?

Rainey (2014): Compression and Conditional Effects

True DGP has no interaction. Without a product term, the model cannot represent this.



Effect Heterogeneity in GLMs.

In non-linear GLMs, robustness to effect heterogeneity is really two questions (or two different sets of assumptions).

1. Is the functional form as estimated adequate (i.e., is a product term needed if not already included)?
2. Do the effects interestingly by subgroup?

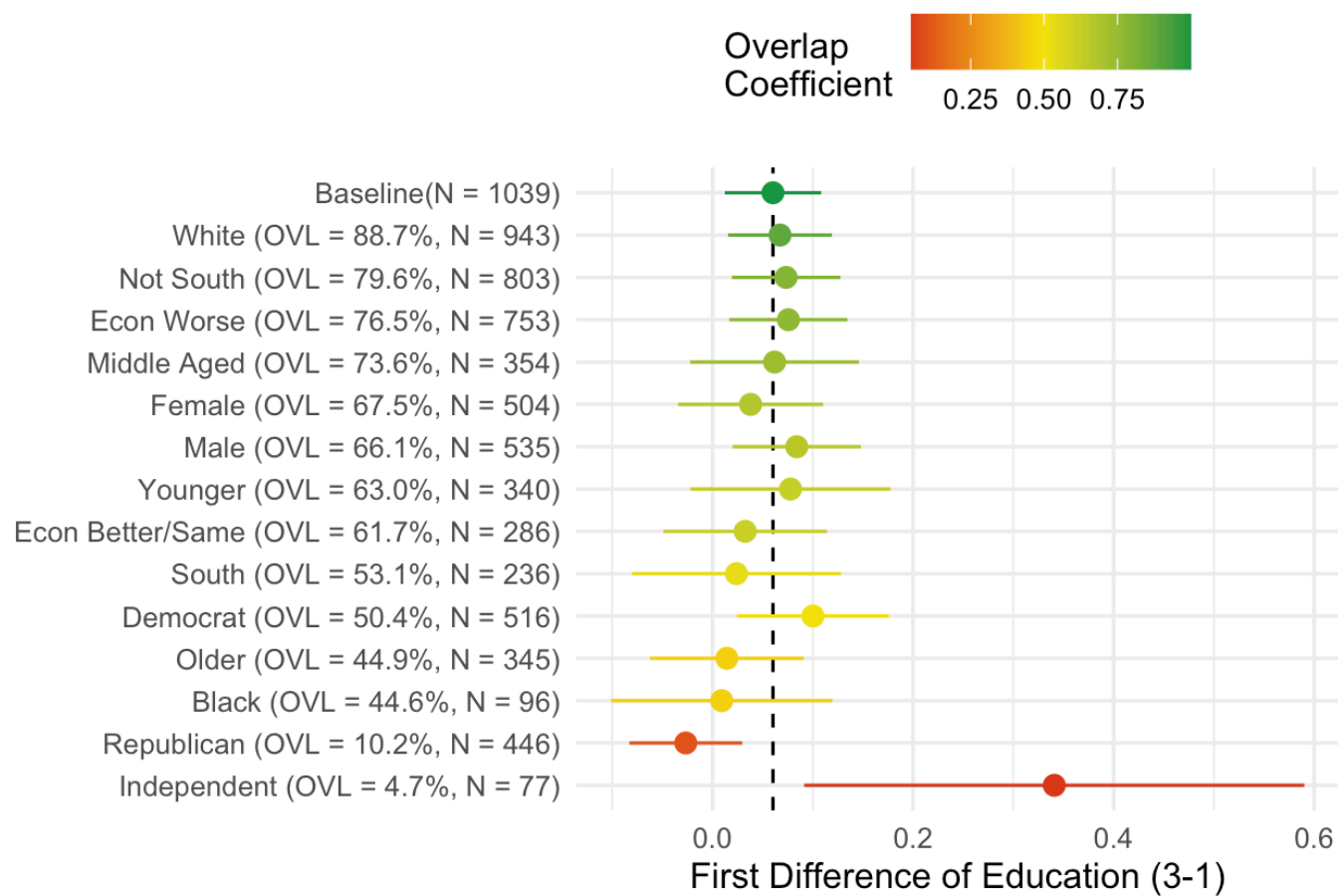
Robustness Evaluation

Instability Type	Interpretation
$\overline{ME}_x^{(a)} \not\approx \overline{ME}_x^{(a)}(z_0)$	Population effect is not representative of subgroup z_0
$\overline{ME}_x^{(a)}(z_0) \not\approx \overline{ME}_x^{(p)}(z_0)$	Conclusions depend on whether moderation is explicitly modeled
$\overline{ME}_x^{(a)} \not\approx \overline{ME}_x^{(p)}(z_0)$	Overall substantive conclusion is not robust

Robustness Workflow

1. Estimate the baseline model.
 - Compute overall effect (average first difference or first derivative)
2. Identify plausible subgroups
3. Estimate one or more alternative specifications
 - Single interaction effect
 - Full interaction effect
 - Subgroup Models
4. Compare baseline and subgroup effects.
 - Evaluate robustness statistics.
 - Evaluate changes in substantive conclusions.
 - Consider whether any observed instability is larger than expected.

Subgroup Robustness Plot





Simulation for Expectations

A question that always comes up - what would we expect under the baseline model specification?

- Here, we would do something similar to before, but simulating the response is slightly different.
1. Estimate the baseline model.
 2. Simulate η on the link scale with $\mathbf{Xb}$.
 3. Transform η into y using the appropriate `r*()` function in R.
 4. Re-run the baseline and subgroup models and calculate robustness.

group	obs	sim	pct
Black	0.45	0.39	0.51
White	0.89	0.89	0.39
South	0.53	0.50	0.47
Not South	0.80	0.82	0.37
Male	0.66	0.73	0.29
Female	0.68	0.67	0.40

Showing 1 to 6 of 14 entries

Previous

1

2

3

Next

Separation

In subgroup analyses, as samples get small, separation problems are more likely. In these cases, finite maximum likelihood estimates may not exist.

If some linear combination of the predictors perfectly predicts the outcome,

$$P(Y = 1 \mid \mathbf{x}) = \begin{cases} 1 & \text{for one region of the predictor space} \\ 0 & \text{for another} \end{cases}$$

then the likelihood can always be increased by making one or more coefficients arbitrarily large.

Consequences of Separation

- Coefficient estimates diverge toward $\pm\infty$
- Standard errors become enormous
- Wald confidence intervals are meaningless
- The model often fails to converge
- Predictions may be numerically unstable

Complete separation

Every success lies on one side of a threshold and every failure on the other.

Separation becomes increasingly common with small samples, rare events, and many indicator variables.

Quasi-complete separation

Only a few observations violate the perfect split, but the likelihood surface is still nearly flat.

Firth's Logistic Regression

Instead of maximizing the ordinary likelihood,

$$\ell(\beta),$$

Firth (1993) maximizes the *penalized* likelihood

$$\ell^*(\beta) = \ell(\beta) + \frac{1}{2} \log |I(\beta)|,$$

where $I(\beta)$ is the Fisher information matrix.

Intuition

The penalty acts like a weak amount of prior information that shrinks estimates away from infinity.

As a result,

- finite coefficient estimates always exist
- first-order small-sample bias is removed
- confidence intervals behave much better
- predictions are typically more stable than ordinary MLE

When should you use it?

- Separation or quasi-separation
- Rare-event logistic regression
- Small samples
- Models with many dummy variables relative to the sample size

Firth's Model in R

The `logistf` package in R has a function `logistf` that takes as input either:

- a glm object, or
- formula and data

The `marginaleffects` functions work with `logistf` models.

- If you see large parameters and really large standard errors in the model output, consider switching to `logistf`.

Banda and Casesse Replication

Banda and Casesse look at the effect of sexism and resentment moderated by party ID on vote in the 2016 election. They initially estimate these effects through an interaction.

1. What do the second differences look like?
2. What do the effects of sexism and resentment look like in the no product term, single interaction, full interaction and subgroup models with pid?
3. Do the moderated effects persist for men and women? What about those who thought the economy got worse vs those who didn't?
 - `econanx` 1 and 2 are worse, 3 is same, 4 and 5 are better.
4. Are any of these robustness statistics much lower than we would expect?

Robustness and Effect Heterogeneity in GLMs

Approach	Assumption
No product term	Link-induced conditionality is sufficient
Single interaction	One effect may vary across groups
Full interaction	All effects may vary across groups
Subgroup Analyses	Coefficients and error structure may differ across groups